

Reproductive Health

REPRODUCTIVE HEALTH : PROBLEMS AND STRATEGIES

Definition

According to WHO, reproductive health means a total well-being in all aspects of reproduction i.e. physical, emotional, social and behavioral.

India was amongst first countries in the world to initiate action plans to attain reproductive health such as family planning programmes (FPP) in 1951

Helping adolescents by providing information about reproductive organs, hygienic sexual practices and STIs

Introduction of sex education in schools

Importance of breastfeeding

Creating awareness about care of pregnant mother, postnatal care of mother and child

Awareness regarding social evils such as sex-abuse and sex related crimes

Discouraging children from believing in myths and having misconceptions about sex-related aspects.

Reproductive and Child Health care (RCH) programmes

Importance of equal opportunities for ♂ and ♀ child

Role of NGOs, audio-visual and print media in creating awareness regarding sex related aspects

Educating people about birth control options

Benefits of massive child immunisation

Benefits of families with small size

POPULATION STABILISATION

According to the 2011 census,
our population growth rate
was less than 2 percent i.e.
20/1000/year

Year	World Population	Indian Population
1900	2 billion	350 million
1900	6 billion	1 billion
1900	7.2 billion	1.2 billion

Reasons For Increase In Population Size:

- Decline in death rate
- Rapid decline in maternal mortality rate (MMR)
- Decrease in infant mortality rate (IMR)
- Increase in number of people in reproductive age
- Increase in health facilities

Measures Taken By Govern- ment To Check Population

- Motivate smaller families by using various contraceptive methods with slogans "Hum do Hamare do", advertisements and posters
- Urban couples adopting: "One child norm"
- Statutory raising of marriageable age:
Female to 18 years
Male to 21 years
- Incentives given to couples with small families.

Features of an ideal contraceptive:

- User-friendly
- Easily available
- Effective
- Reversible
- No/least side-effects
- No interference with libido or act of coitus

BIRTH CONTROL/ CONTRACEPTION

NATURAL METHODS

- Principle of avoiding physical meeting of the egg and sperms
- Chances of failure are high

There are two principle n.ethods of birth control:

- User-friendly
- Easily available
- Effective
- Reversible
- No/least side-effects
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NATURAL METHODS

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- Chances of failure are high

Method

Mode of Action (MoA)

Periodic abstinence	- Couples abstain from coitus from day 10 to 17 of the menstrual cycle i.e. fertile period
Withdrawal method/ Coitus interruptus	- Insemination is avoided as the male partner withdraws his penis from the vagina just prior to) ejaculation
Lactational amenorrhea	- Absence of menstruation upto 6 months during period of intense lactation following parturition

ARTIFICIAL METHODS



The various methods of contraception are enlisted below:

- I. Barrier methods
- II. Spermicidal jellies
- III. IUDs
- IV. Oral pills
- V. Injections and implants
- VI. Emergency contraceptives
- VII. Surgical methods

(a) Tubectomy (in females)

(b) Vasectomy (in males)

ARTIFICIAL METHODS

Barrier methods

Prevent ovum and sperm from physically meeting
Self inserted and offer privacy to user

Condoms & its types

Made up of rubber and thin latex

Parameter

Region covered

Provides protection from STIs

Males

Penis

Yes

Male condom (Nirodh)

Females

Vagina and cervix

Yes

Female condom

Diaphragms, cervical caps,
vaults

- Rubber barriers that cover the cervix during coitus
- Reusable
- Do not protect from STIs
- Used by females only

Spermicidal jellies, foams
and creams

- Kill the sperms by creating acidic pH
- Used along with barrier methods to increase their efficiency

Implants

Placed under skin

Effective periods are much longer

- Composition : Progestogens alone
- /Combination of Progestogens and Estrogens-
- Mode of Action (MoA)
Inhibit ovulation and implantation
- After the quality of cervical mucus to retard
 - entry of sperms
 - Injections usually share similar MoA and composition as implants

Intra-uterine devices (IUDs)

- Inserted by doctors or expert nurses in uterus
- IUDs are one of the most widely accepted methods of contraception in India.

Types

- Non medicated IUDs
- Example : "Lippes loop"
- Medicated IUDs

Hormone releasing IUDs

Progestasert, LNG-20

- Make the uterus unsuitable for implantation and the cervix hostile to the sperm

Copper coated IUDs

Cu7, Multiload 375

- Increase phagocytosis of sperms within uterus
- Cu ions suppress sperm motility and fertilizing capacity of sperms

Oral Contraceptive Pills (OCP) or tablets

Parameter	Non-steroidal	Steroidal
Example or composition	SAHELI	rogestogens (Prg) alone or combination of Prg and Estrogens (Est)
Mode of action	Interferes with implantation	Inhibit ovulation and implantation; also alter the quality of cervical mucus to retard entry of sperms
Dosage	'Once a week' pill	Pills have to be taken daily for a period of 21 days starting preferably within first 5 days of menstrual cycle
Effectiveness	High contraceptive value with very few	Pills are very effective with lesser side effects and well accepted by females

Saheli was developed at CDRI, Lucknow, Uttar Pradesh

Emergency contraceptives

Types

Progestogens alone
Combination of Prg + Est

IUDs

Characteristics

Effective with 72 hrs of coitus
Used to prevent conception resulting from
rape or unprotected intercourse

Surgical/Sterilisation methods

Poor reversibility but highly effective

Mode of action

Blocks gamete transport

Types

Tubectomy

In females

Cut and tie fallopian tubes
Incision in abdomen or
through vagina

Vasectomy

In males

Cut and tie vas deferens

Small incision on the scrotum

MEDICAL TERMINATION OF PREGNANCY (MTP)/INDUCED ABORTION

MTP : Intentional or voluntary termination of pregnancy before full term
MTP was legalised in India in 1971

When can MTP be performed?

- Unwanted pregnancy due to rape, failure of contraception, casual unprotected intercourse
- If continuation of pregnancy could harm the mother or foetus or both
- **Permission of how many medical practitioners is needed for MTP depending on duration of gestation?**
Less than/upto 12 week
More than 12 but less than 24 weeks

Intention behind MTP amendment act 2017, (GoI)

- Reducing the incidence of illegal abortion
- Decrease consequent maternal mortality and morbidity
- MTPs are safe upto 12 weeks but riskier in 2nd trimester yet both are legal
- Amniocentesis and MTPs have been misused in context of female foeticide

AMNIOCENTESIS

- Analyse foetal cells and dissolved substances from amniotic fluids
- Technique used to check for genetic disorders such as Down's syndrome, hemophilia, sickle-cell anemia etc.
- Statutory ban on this technique in India to prevent female foeticide.

SEXUALLY TRANSMITTED INFECTIONS (STIs)

- Alternately named: Venereal diseases (VD) or reproductive tract infections (RTIs)
- High vulnerability/risk group: 15-24 years
- Mode of transmission (MoT): Sexual intercourse

Category	Disease
Bacterial	Gonorrhea, Syphilis, Chlamydia
Protozoan	Trichomoniasis
Viral	Genital herpes, Hepatitis-B, Genital warts, AIDS

- Bacterial and protozoan diseases are completely curable if detected early and treated properly
- Other MoT for hepatitis-B virus and HIV infection include:
 - Sharing of injection needles, surgical instruments with infected persons
 - Transfusion of blood
 - From infected mother to foetus

INFERTILITY

- Infertile couple: Unable to produce children inspite of 2 years of unprotected sexual co-habitation
- **Reasons for infertility:**
 - Physical
 - Congenital
 - Diseases
 - Immunological
 - Psychological
- Infertility as a problem could be with either the male or female partner.
- In India, female is blamed often than male for the couple being childless

Help For Infertile Couples Comes In The Form of ASSISTED REPRODUCTIVE TECHNOLOGIES (ART)

- Alternately named: Venereal diseases (VD) or reproductive tract infections (RTIs)
- High vulnerability/risk group: 15-24 years
- Mode of transmission (MoT): Sexual intercourse

Parameter	in-vitro fertilisation	in-vivo fertilisation
▪ Site of fertilisation	▪ Outside the body in simulated, conditions in laboratory	▪ In the female, reproductive tract
▪ Can female produce ova?	▪ Yes	▪ No
▪ Embryo transfer	▪ Yes	▪ No
▪ Example of techniques	▪ ZIFT, IUT, ICSI	▪ GIFT, AI, IUI

Site of Embryo Transfer (ET) based on number of blastomeres

Parameter

- Location
- Technique

Upto 8 blastomeres

in fallopian tube

ZIFT: Zygote Intra fallopian transfer

More than 8 blastomeres

in uterus

- IUT: Intra uterine transfer
- Other details of ART involved:
- ICSI : Intra cytoplasmic sperm injection
- Sperm injected directly into the egg
- Artificial Insemination (AI):
- Semen introduced in vagina or uterus
- Low sperm count or inability of male to inseminate female
- IUI : Intra uterine insemination
- GIFT: Gamete intra fallopian transfer
- Female can provide conditions for fertilisation and further development

Test tube baby programme
involves techniques with *in-vitro fertilisation*

Early detection

Symptoms

Itching, fluid discharge, slight pain, swellings in the genital region

Late detection

Complications

Pelvic inflammatory diseases (PIDs), abortions, still births, ectopic pregnancies, infertility, cancer of repro-

Preventive measures to avoid STIs:

- Avoid sex with unknown partners/- multiple partners
- Always try to use condoms during coitus